N100 Financial Intelligence Platform

Analyst Guide

Streamlit Dashboard • FastAPI • Financial Analytics • Reports

Day 44 Documentation

1. Platform Overview

The N100 Financial Intelligence Platform is designed for analysts who need a single workflow for exploring company fundamentals, comparing peers, screening the Nifty 100 universe, reviewing trends, and generating reports.

The project plan defines an eight-screen Streamlit dashboard, a FastAPI service under /api/v1, a SQLite database, financial KPI calculations, data-quality validation, clustering, valuation, and reporting components.

- Use the dashboard for interactive exploration.
- Use the API for programmatic access and integration.
- Use reports/tearsheets/ for company PDF outputs.
- Use docs/openapi.json for the API contract.

2. Starting the Platform

Activate the project virtual environment before running commands. The project database is stored in database/n100.db.

For a normal analyst session, start FastAPI first and Streamlit second. This keeps the API available for health checks and integrations while the dashboard is running.

- PowerShell: .venv\Scripts\Activate.ps1
- FastAPI: uvicorn src.api.main:app --host 127.0.0.1 --port 8000
- Streamlit: streamlit run src\dashboard\app.py --server.port 8501
- API docs: http://127.0.0.1:8000/docs
- Dashboard: http://127.0.0.1:8501

3. Home Screen

The Home screen is the starting point for the dashboard. It provides the analyst with an overview of the company universe and high-level financial information.

Use the Home screen when you want to understand the overall dataset before drilling into an individual company, sector, peer group, or screening strategy.

- Review headline KPIs.
- Review sector distribution.
- Identify high-quality companies for deeper analysis.
- Use the navigation menu to move to the other screens.

4. Company Profile Screen

The Company Profile screen is intended for company-level research. Select a ticker such as TCS, INFY, HCLTECH, BEL, or HAL and review the available company information and financial indicators.

The profile workflow is also the basis of the Day 43 performance test. Five representative tickers were tested and each profile loaded well below the three-second target.

- Select the company/ticker.
- Review company and sector information.
- Review financial KPIs and ratios.
- Inspect revenue, profit and ratio trends.
- Review available pros/cons and related analyst information.

5. Streamlit Screener

The Screener is the main interactive filtering tool. It allows an analyst to combine quality, growth, leverage, cash-flow, valuation and dividend constraints.

A common example is an ROE minimum of 15%. The API integration test uses the same core condition through GET /api/v1/screener?min_roe=15.

- ROE minimum
- Debt/Equity maximum
- FCF minimum
- Revenue CAGR minimum
- PAT CAGR minimum
- OPM minimum
- P/E maximum
- P/B maximum
- Dividend Yield minimum
- ICR minimum

6. Peer, Trend and Sector Screens

Peer Comparison is used for relative analysis inside a peer group. Trend Analysis focuses on historical financial movement. Sector Analysis aggregates companies at the sector level and helps identify sector medians and relative performance.

Use these screens after the Screener or Company Profile has identified a company or group that needs context.

- Peer Comparison: compare company KPIs against peers.
- Trend Analysis: inspect multi-year financial trends.
- Sector Analysis: compare sector-level metrics and companies.

7. Capital Allocation and Annual Reports

The Capital Allocation screen presents capital-allocation patterns and related company information. The Annual Reports screen helps analysts locate company report links and review available reporting years.

These screens are useful when fundamental ratios need to be supplemented by capital-allocation or source-document context.

- Use Capital Allocation for allocation-pattern analysis.
- Use Annual Reports to locate source documents.
- Cross-check conclusions against primary company disclosures when appropriate.

8. PDF Tearsheets

Company tearsheets are generated by the reporting layer and stored under reports/tearsheets/. The project plan explicitly requires company PDF tearsheets and an analyst guide covering their generation.

A typical workflow is to choose a company, invoke the project reporting function or report screen, generate the PDF, then review the resulting file before sharing it.

- Select a target company.
- Generate the company tearsheet using the reporting workflow.
- Open the generated PDF from reports/tearsheets/.
- Check that the company name, metrics, charts and pages are correct.
- For final sign-off, verify file size and inspect sampled PDFs for text overflow.

9. FastAPI Usage

The FastAPI service provides programmatic access to company, screener, sector, peer, valuation, portfolio, document and health modules. The service is mounted below /api/v1.

The project plan specifies health, company, screener and sector tests and requires an OpenAPI specification.

- Health: curl http://127.0.0.1:8000/api/v1/health
- Companies: curl http://127.0.0.1:8000/api/v1/companies
- TCS: curl http://127.0.0.1:8000/api/v1/companies/TCS
- Screener: curl "http://127.0.0.1:8000/api/v1/screener?min_roe=15"
- Sectors: curl http://127.0.0.1:8000/api/v1/sectors

10. Testing and Performance

The project includes ETL, KPI, data-quality, API, integration and performance tests. Day 42 requires at least 60 collected tests with zero failures. Day 43 adds concurrent API, dashboard performance and end-to-end service checks.

Recorded Day 43 results were 10/10 successful concurrent API requests in 0.1099 seconds wall time, average API response 0.0860 seconds, and an average Company Profile load of 0.0499 seconds across five tickers.

- Run all tests: pytest tests\ -v
- Generate HTML report: pytest tests\ --html=reports\pytest_report.html
- Run performance tests: pytest tests\performance\ -v -s
- Investigate failed tests before considering the platform ready for sign-off.

11. Database Optimisation

The SQLite optimisation pass created or verified indexes on large tables using company_id and year access paths. SQLite ANALYZE was also executed.

The indexed tables include profit/loss, balance sheet, cash flow, market cap, financial ratios, historical ratios, peer groups, pros/cons, sectors and stock prices.

- Use PRAGMA index_list(table_name) to inspect indexes.
- Keep the database file at database/n100.db.
- Re-run ANALYZE after substantial data changes when appropriate.

12. Troubleshooting and Analyst Checklist

If a service does not start, first check the terminal for import errors and verify the database path. If a port is already in use, identify the process before restarting.

Before distributing analysis, verify the selected company/ticker, filter configuration, date/year context, and generated report files.

- Port 8000: netstat -ano | findstr :8000
- Port 8501: netstat -ano | findstr :8501
- Database: confirm database/n100.db exists.
- Tests: pytest tests\ -v
- API docs: http://127.0.0.1:8000/docs
- Tearsheet directory: reports/tearsheets/